

Step 4 DataBase 2 Project

ArtConnectApp

Brief Description of the architecture:

The key idea is that each layer only talks to the one directly below it:

UI → Service → DAO → Database

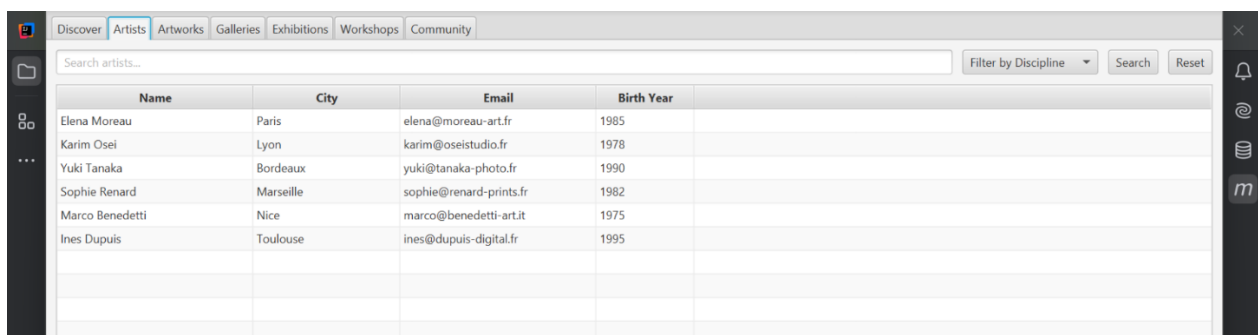
UI (User Interface) The screens the user sees and interacts with. It just displays data and captures actions (clicks, form inputs). It doesn't know anything about the database.

Service Layer The "brain" of the app. It contains the business rules, for example "is this artwork available before buying it?" It receives requests from the UI and decides what to do.

DAO (Data Access Object) The only layer that talks to the database. It contains all the SQL queries (SELECT, INSERT, UPDATE, DELETE). Nothing else in the app writes SQL.

Database (MySQL) Where all the data actually lives permanently. Without this, everything resets when you close the app.

And here as we can see below the database was linked successfully by looking at both the SQL inserts and the UI that we have:



The screenshot shows the ArtConnectApp interface with the 'Artists' tab selected. A table displays the following data:

Name	City	Email	Birth Year
Elena Moreau	Paris	elena@moreau-art.fr	1985
Karim Osei	Lyon	karim@oseistudio.fr	1978
Yuki Tanaka	Bordeaux	yuki@tanaka-photo.fr	1990
Sophie Renard	Marseille	sophie@renard-prints.fr	1982
Marco Benedetti	Nice	marco@benedetti-art.it	1975
Ines Dupuis	Toulouse	ines@dupuis-digital.fr	1995

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INSERT INTO CommunityMember
```

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(MemberID, CMName, CMEmail, CMCity, CMBirthyear, CMPhone, MembershipType)
```

```
VALUES
```

```
(1, 'Alice Fontaine', 'alice@mail.fr', 'Paris', '1988-04-02', '+33 6 01 11 22 33', 'Premium'),  
(2, 'Bruno Sauvage', 'bruno@mail.fr', 'Lyon', '1975-08-19', '+33 6 02 11 22 33', 'Standard'),  
(3, 'Céline Morel', 'celine@mail.fr', 'Bordeaux', '1992-12-25', '+33 6 03 11 22 33', 'Premium'),  
(4, 'David Kante', 'david@mail.fr', 'Marseille', '1983-03-07', '+33 6 04 11 22 33', 'Basic'),  
(5, 'Emma Leclaire', 'emma@mail.fr', 'Nice', '1997-06-14', '+33 6 05 11 22 33', 'Standard'),  
(6, 'Félix Garnier', 'felix@mail.fr', 'Toulouse', '1969-11-30', '+33 6 06 11 22 33', 'Premium'),  
(7, 'Gaëlle Perrin', 'gaelle@mail.fr', 'Paris', '2001-01-08', '+33 6 07 11 22 33', 'Basic'),  
(8, 'Hugo Descartes', 'hugo@mail.fr', 'Lyon', '1986-09-21', '+33 6 08 11 22 33', 'Standard');
```